

Your grade: 100%

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Reveal Solution

1. Maya is building a chat-based AI assistant that needs to answer questions such as "What is the weather like in a location?" and "How many customers do I have?" She decides to use the Model Context Protocol (MCP) so the assistant can work with different data sources. How should her system be structured so the assistant can discover and use the correct tools?
- ☐ The MCP host contains the MCP client, which connects to one or more MCP servers that provide tools linked to databases, APIs, or local files.
- ☐ The LLM connects directly to every database and API so that MCP does not need to manage tools or server connections.
- ☐ The MCP server runs inside the LLM so that tools do not need to be discovered or executed through an MCP client.
- ☐ The MCP client can only connect to one server, so all tools must be hosted in a single MCP server instance.
- ☒ **Nice work**
- The MCP host includes the client, which connects to MCP servers that expose tools calling databases, APIs, or local data sources.
2. Maya is designing an AI system for an enterprise organization that needs to connect to internal CRM, ERP, inventory APIs, and a database of customer records. She wants her assistant to use data obtained having to rewrite integrations for every new project. Which benefit of the Model Context Protocol (MCP) best supports her goal?
- ☐ UI automation, because MCP automatically builds front-end interfaces for business tools.
- ☐ Data storage, because MCP includes built-in databases to store enterprise information.
- ☒ **Standardized integration, because MCP provides a common protocol that eliminates the need to build custom connectors for each tool.**
- ☐ Hardware provisioning, because MCP installs and manages infrastructure for AI environments.
- ☒ **Nice work**
- MCP's standardized integration allows developers to connect models to external data and services using a common protocol, reducing the need for custom, one-off integrations and speeding up development.
3. Which of the following is NOT a capability of an MCP server?
- ☐ Providing tools, resources, and predefined prompts for AI clients.
- ☒ **Automatically replacing human developers for all coding tasks.**
- ☐ Enabling AI agents to handle routine tasks like support ticket updates or repository management.
- ☐ Connecting to external APIs and systems to fetch data or perform actions.
- ☒ **Nice work**
- MCP assists AI in interacting with systems but does not replace human developers entirely.
4. Jordan is building an AI assistant that needs to interact with multiple external services, such as a calendar API, a weather API, and a database of customer records. She wants her assistant to use data obtained having to rewrite integrations for every new project. Which approach should Jordan use?
- ☐ She should only connect to one external service and configure all tools there.
- ☐ She should replace all APIs and connect the LLM directly to each external service manually.
- ☐ She should store all external data locally on the AI assistant device and bypass servers.
- ☒ **She should use MCP so the AI assistant can access tools and data from MCP servers via a standard protocol.**
- ☒ **Nice work**
- MCP allows AI agents to retrieve context and call tools in a uniform way through MCP clients and servers.
5. Alex, a Python developer, needs to connect to an MCP server that provides FastMCP tools for library documentation. He wants to choose the transport that allows communication with a server running locally on his machine without using network protocols. Which approach should Alex use?
- ☐ Use HTTP transport with the MCP server's URL endpoint.
- ☒ **Use STDIO transport to connect via Standard Input/Output.**
- ☐ Use Server-Sent Events (SSE) transport to connect over the internet.
- ☐ Use In-Memory transport to communicate with remote servers over HTTP.
- ☒ **Nice work**
- STDIO transport communicates with MCP servers running locally, using STDIN and STDOUT without requiring a network connection.
6. Mike is developing an AI application that integrates multiple MCP servers, one providing LLM-optimized library documentation and another offering access to the integration library of API calls. Jordan wants the agent to remember previous conversations across multiple user prompts and handle multiple server connections simultaneously. Which setup should Jordan implement?
- ☐ Use a single MCP server with STDIO transport and no memory management.
- ☒ **Use MultiServerMCPClient to connect to both HTTP and STDIO MCP servers, and implement an InMemoryServer for conversation memory.**
- ☐ Use STDIO transport for remote servers and do not use asynchronous execution.
- ☐ Use HTTP transport only and fetch tools separately for each server without a client manager.
- ☒ **Nice work**
- MultiServerMCPClient allows simultaneous connections to multiple servers, while InMemoryServer lets the agent maintain context across interactions.
7. In the hands-on lab, which two tools are exposed by Context? to AI IDEs after MCP integration?
- ☐ analyzeCode and generatePrompt.
- ☐ findFrameworks and readAgent.
- ☐ fetchAPIKey and validateDocs.
- ☒ **resolveLibraryId and getLibraryDocs.**
- ☒ **Nice work**
- Context exposes 'resolveLibraryId' to identify the relevant framework or library and 'getLibraryDocs' to retrieve structured documentation and code snippets for AI-assisted development.